

Tengrui (Rio) Kong

Rolla, MO | kongtengrui123@gmail.com | github.com/Riokong | linkedin.com/in/teng-rui

RESEARCH INTERESTS

Trustworthy machine learning for high-stakes scientific and biomedical domains. My research focuses on graph-based deep learning for neuroimaging, foundation model evaluation in data-scarce settings, LLM robustness and faithfulness, and AI-assisted scientific discovery.

EDUCATION

Incoming Ph.D. Student in Computer Science Missouri University of Science and Technology - Rolla, MO Research focus: machine learning for biomedical and bioinformatics applications.	Starting Aug 2026 Advisor: Dr. Seung-Jong Park
M.S. in Computer Science Missouri University of Science and Technology - Rolla, MO Selected coursework: Deep Learning, Natural Language Processing, Machine Learning, Advanced Topics in LLMs, Advanced Topics in Computer Vision, Analysis of Algorithms, Probability & Applications in Computing, Foundations of Data Management.	Aug 2024 - May 2026 GPA: 3.81 / 4.0
M.S. in Economics and Finance University of International Business and Economics - Beijing, China	2015
B.S. in Communication Engineering East China Jiaotong University - Nanchang, China	2012

PUBLICATIONS AND MANUSCRIPTS

- [1] Tengrui Kong, Tao Wang, Minheng Chen, Ping Ma, Dajiang Zhu, Xiaowei Yu.
Are Foundation Models Enough for Data-Scarce Functional Connectivity Diagnosis? Revisiting Task-Specific Learning with BrainLRR. Under review, WACV 2027.
- [2] M. Alizade, Tengrui Kong, Suman K. Maity.
SciTables: A Dataset and Evaluation Framework for Complex Table-to-Text Generation. Under review, VLDB 2026.
Contribution: dataset pipeline construction, LaTeX table extraction, and LLM benchmarking.

SELECTED RESEARCH HIGHLIGHTS

- First-authored BrainLRR, a low-rank regularized graph Transformer for data-scarce functional connectivity diagnosis.
- Benchmarked task-specific models, vision backbones, LLMs, and neuroimaging foundation models across ASD, Alzheimer's disease, MCI, and Parkinson's disease datasets.
- Built reproducible PyTorch and SLURM-based HPC pipelines for multi-seed biomedical machine learning experiments.
- Investigated LLM robustness under politically charged misinformation and presupposition-based prompts.
- Developed LLM-guided optimization methods for astrophysics parameter estimation.

RESEARCH EXPERIENCE

Graduate Research Assistant Missouri S&T, CoBAI Lab - Rolla, MO	Dec 2025 - May 2026 Advisor: Dr. Xiaowei Yu
- First-authored BrainLRR, a large-scale benchmark evaluating task-specific models and foundation models for data-scarce functional connectivity diagnosis across ASD, Alzheimer's disease, MCI, and Parkinson's disease datasets. - Designed and trained a graph-based Transformer with stochastic node masking and low-rank regularization, improving ASD classification AUROC from 75.6% to 79.1% on resting-state fMRI functional connectivity. - Benchmarked neuroimaging foundation models, including BrainLM and BrainSegFounder, as well as vision backbones such as ViT and ResNet, against task-specific baselines including Brain Network Transformer. - Conducted ROI-level interpretability analysis to connect model predictions with clinically meaningful neuroimaging patterns. - Built reproducible HPC training pipelines using PyTorch, SLURM, and the Hellbender cluster. - GPU-accelerated a custom low-rank representation loss using torch.linalg.svd, achieving approximately 10x speedup over CPU-based SciPy SVD.	

SELECTED RESEARCH PROJECTS

LLM-Guided Astrophysics Parameter Optimization	Mar 2026 - May 2026
- Developed an LLM-guided optimizer for the 3ML + HAL HAWC astrophysics pipeline, replacing the Minuit minimizer with structured prompting and finite-difference gradient estimation. - Reduced the log-likelihood gap from 1,760 to 38 after 200 iterations. - Deployed the system locally using Ollama and Llama 3.1 to avoid external API costs and improve reproducibility. - Implemented convergence diagnostics, stall detection, and parameter-bound clipping for robust optimization.	
Political Misinformation Robustness in LLMs	Sep 2025 - Dec 2025
Constructed a benchmark using Wikipedia controversies and fact-checking archives to evaluate LLM robustness under politically charged prompts.	

- Evaluated models including LLaMA-3.3-70B, Qwen-2.5-72B, GPT-OSS-120B, and Aya-Expanse-32B.
- Analyzed model behavior under presupposition-based prompts to study misinformation acceptance and reasoning robustness.

TECHNICAL SKILLS

Machine Learning and Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, graph neural networks, Transformer architectures.

LLMs and NLP: LLM fine-tuning, LoRA, QLoRA, supervised fine-tuning, prompt engineering, LLM-as-a-judge, RAG, BERT, SpaCy, NLTK, Ollama.

Neuroimaging and Biomedical AI: fMRI functional connectivity, nilearn, NetworkX, ABIDE, ADNI, PPMI, CC200 atlas, ROI-level analysis.

Infrastructure: SLURM, HPC, Hellbender cluster, AWS EC2, S3, RDS, Git, multi-GPU training.

Programming: Python, SQL, Java, JavaScript, C++.

INDUSTRY EXPERIENCE

Principal Research Specialist

Nov 2020 - Jun 2024

Changsha Science & Technology Bureau - Changsha, China

- Applied quantitative analysis, Python, and SQL-based modeling to evaluate large-scale R&D portfolios and support technology policy decisions across 200+ companies.
- Designed statistical evaluation frameworks and forecasting dashboards for strategic planning and capital allocation.

Bond Investment Analyst, Fixed Income Department

Jun 2015 - Nov 2020

First State Cinda Fund Management / Yingda Securities - Shenzhen, China

- Applied statistical modeling to credit risk assessment, fixed-income research, and portfolio analysis.
- Produced issuer reports and investment analyses supporting bond portfolio decisions.

CERTIFICATIONS

AWS Certified Cloud Practitioner - Issued Jul 2025; Expires Jul 2028